

# WASP-52b

R-0541

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-52\_240922 · created 2026-07-22T00:26:05+00:00

## BENCHMARK: INACCURATE

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460575.8364 +/- 0.0026 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.1562 +/- 0.0096               |
| Transit depth (Rp/Rs)^2                           | 2.44 +/- 0.3 [%]                |
| Orbital Inclination (inc)                         | 89.65 +/- 1.23                  |
| Airmass coefficient 1 (a1)                        | 0.948 +/- 0.011                 |
| Airmass coefficient 2 (a2)                        | 0.04 +/- 0.012                  |
| Scatter in the residuals of the lightcurve fit is | 1.2 %                           |
| Best Comparison Star                              | #1 - [448, 329]                 |
| Optimal Method                                    | PSF photometry                  |
| Transit Duration (day)                            | 0.0867 +/- 0.0014               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                          |
|--------------------------------------|----------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.1562 $\pm$ 0.0096 vs 0.1646 (deviation 0.86 $\sigma$ ) |
| Duration fit vs published            | 0.0867 d vs 0.0758 d (deviation 7.65 $\sigma$ )          |
| Mid-time O-C                         | 4.3 min                                                  |
| Depth SNR                            | 16.0                                                     |
| Residual scatter                     | 1.2 %                                                    |

### Light curve

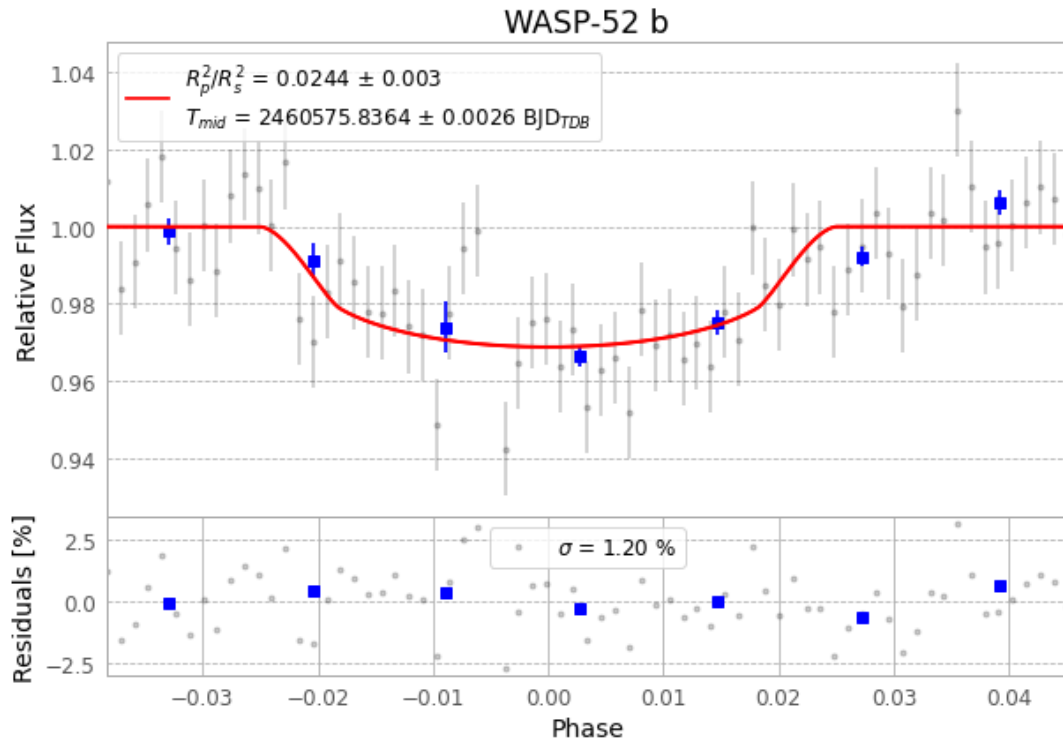


Figure 1 — FinalLightCurve\_WASP-52 b\_2024-09-21.png (SHA-256 810158e6b0e3159d...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [328, 247], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 bf448c87fbc56bf0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit b365494

## Data provenance

71 FITS frames (47 MB), WASP-52240922061815.FITS → WASP-52240922094816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-240922.log (7f9641b6b104...), FinalLightCurve\_WASP-52 b\_2024-09-21.png (810158e6b0e3...), FinalLightCurve\_WASP-52 b\_2024-09-21.csv (b7e19e915818...)

## Compute resources

Wall time 105.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-22 00:26Z.